INCH-POUND

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# DETAIL SPECIFICATION

# TEST STAND, HYDRAULIC SYSTEM COMPONENTS

MIL-DTL-87976A is inactive for new design and is no longer used, except for replacement purposes.

## 1. SCOPE

1.1 Scope. This document covers the requirements for a non-portable test stand that will test aircraft hydraulic components.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this detailed specification. This section does not include documents cited in other sections of this detailed specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 and 4 of this detailed specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, Bldg 28, 2145 Monahan Way, Wright-Patterson AFB OH 45433-7017 or emailed to engineering.standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-F-27656 | Filter Unit , Fluid, Pressure MXU-408/M, 5 Micron, Hydraulic. |
| MIL-PRF-5606 | Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance |
| MIL-PRF-6083 | Hydraulic Fluid, Petroleum Base, for Preservation and Operation |

DEPARTMENT OF DEFENSE STANDARDS

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-882 | System Safety Program Requirements. |
| MIL-STD-2073-1 | Standard Practice for Military Packaging |
| MS21919 | Clamp, Loop-Type, Cushioned, Support. |

DEPARTMENT OF DEFENSE HANDBOOK

| | |
|---|---|
| MIL-HDBK-454 | General Guidelines for Electronic Equipment |
| MIL-HDBK-470 | Designing and Developing Maintainable Products and Systems, Volume I |
| MIL-HDBK-808 | Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment |

AIR FORCE - NAVY AERONAUTICAL

| | |
|---|---|
| AND10059 | Fitting - Style of Pipe Threaded Flared Tube and Hose. |
| AND10104 | Tubing, Steel, Corrosion Resistant, Round Standard Dimensions for. |
| AND20001 | Drive-Type XI Engine Accessory. |

(Copies of these documents are available online at https://assist.dla.mil.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM D 3951 | Standard Practice for Commercial Packaging. |



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(Copies of these documents are available from www.astm.org.

AMERICAN STANDARDS

C-50 Rotating Electrical Machinery.

(Copies of these documents are available from www.ansi.org.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

Standard MG-1 Motors and Generators.

(Copies of these documents are available from www.nema.org.

NATIONAL FIRE PROTECTION ASSOCIATION

Code 70 National Electrical Code (Electric Wiring Apparatus).

(Copies of these documents are available from www.nfpa.org.

SAE INTERNATIONAL

| | |
|---|---|
| SAE-AMS-STD-595 | Colors Used in Government Procurement |
| SAE-AS4395 | Tube Connection, Fitting End - Flared, Design Standard |
| SAE-AS5440 | Hydraulic Systems, Aircraft, Design and Installation Requirements For |
| SAE-AS33611 | Tube Bend Radii |

(Copies of this document are available from or www.sae.org.

2.2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification needs to take precedence.



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3. REQUIREMENTS

3.1 General. The requirements specified herein are minimum; items providing additional characteristics, greater accuracy and improved performance will be acceptable.

3.2 General requirements. The requirements of SAE-AS5440 for materials, fluids, reservoir venting, tubing, and cleaning of parts and systems apply as requirements of this specification with the exceptions and additions specified herein. This test stand shall be designed for the performance testing of hydraulic system components such as valves, actuating cylinders, solenoid valves, reservoirs, accumulators, pumps, etc., using either MIL-PRF-5606 or MIL-PRF-6083 hydraulic fluids at an operating temperature of 100 to 160 degrees Fahrenheit (°F). The test stand shall contain the following basic flow and pressure circuits or capabilities:

a. A pump test circuit suitable for testing fixed and variable volume pumps with displacements up to 30 gallons per minute (gpm), discharge pressures up to 3200 pounds per square inch gauge (psig), inlet pressures from 0 to 150 psig, and case drain flows from 0.13 to 1.66 gpm at pressures from 0 to 160 psig.

b. A 500 to 5000 rpm variable speed drive which shall deliver sufficient torque to drive a hydraulic pump with a displacement of 30 gpm at 3750 rpm and at a pressure of 3000 psig.

c. A hydraulic static pressure circuit capable of generating hydrostatic pressures up to 25,000 pounds per square inch (psi).

d. A pneumatic static pressure test circuit capable of providing pressures up to 3,000 psig.

e. A dynamic test circuit: capable of testing a variety of aircraft components such as actuating cylinders, relief valves, selector valves, shuttle valves, etc., at flow rates up to 30 gpm with pressures up to 3750 psig and back pressures up to 200 psig. The system shall be capable of generating 5000 psig pressures at reduced flow rates.

f. Direct current (DC) and alternating current (AC) electrical power supplies.

g. Measuring instruments and circuits shall be provided so that the above can be measured throughout their various ranges.

3.2.1 Cabinet. The test stand equipment shall be located within a two section cabinet. One section shall contain the drive console, and the other section shall contain the control console. In use the two sections are to be fastened together. All hardware required to make this connection shall be contractor furnished.

3.2.1.1 Drive console cabinet. The drive console cabinet shall contain the variable speed drive, pump test mounting pad, panel, hydraulic and pneumatic ports, high pressure filter, ripple filter, sink, shield, main system hydraulic pump with electric motor, pump volume and pressure controls, and necessary electrical components.



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3.2.1.2 Control console cabinet. The control console cabinet shall contain controls including the variable speed drive controls, indicating instruments, electrical power supplies, hydraulic reservoir, hydraulic and pneumatic circuit, panel, sink, writing tabletop, sink type storage and other necessary components as required herein.

3.2.1.2.1 Writing top and storage. The control console cabinet shall include a 21x34 inch (minimum) writing tabletop coated with a material resistant to damage from dropped wrenches, scratching, etc. The tabletop shall be hinged to provide access to the deep sink type storage area which shall also be provided. The storage area shall be 17x30x8 inch deep (minimum) and shall be provided with a drain connection.

3.3 Hydraulic and pneumatic systems. The test stand shall include hydraulic and pneumatic systems to comply with requirements specified herein.

3.3.1 Hydraulic and pneumatic circuits. The test stand's hydraulic and pneumatic system circuits shall conform to the schematic shown on Figure 1. The test stand's actual hydraulic and pneumatic system schematic shall be included on a suitably engraved metal sheet and installed on the main panel at a position where it can be read easily.

3.3.2 Plumbing. The test stand's fluid lines and fittings shall be of sufficient size and strength and shall be fabricated from materials sufficient to meet requirements specified herein.

3.3.2.1 Tubing. All low pressure or suction tubing used in hydraulic oil circuits may be aluminum alloy tubing. All high pressure tubing used in the hydraulic circuits shall be annealed, stainless steel seamless tubing. The minimum radii of tubing bends shall be in accordance with SAE-AS33611.

3.3.2.1.1 Tubing supports. Tubing or pipe shall not be welded to supports. All supports shall be of the clamp type conforming to the requirements of MS21919.

3.3.2.1.2 Tubing dimensions. Tubing dimensions shall conform to AND10104. Tubing in the 5000 psi circuits shall be designed to 20,000 psi burst pressure. Tubing in the 25,000 psi circuit shall be designed to 75,000 psi burst pressure. Return lines shall be designed to burst pressure equal to 400 percent of the maximum operating pressure. The cross sectional area of the tubing shall be such that cavitation, turbulence, or undue-temperatures rise will not be developed in the system. Tube sizes shall be such that the maximum fluid velocity shall not exceed 18 feet per second in operating valves and lines. Velocities in return and suction lines shall not exceed five feet per second. The maximum-fluid velocity through any instrument shall be as required for proper operation. The suction and return lines shall be designed for 125 psi maximum working pressure. Valve size to be compatible with tube size.

3.3.2.1.3 Fittings. All high pressure line fittings and panel mounted bulkhead fittings shall be of the flared tube type conforming to standard AND10059. High pressure fittings shall be corrosion resistant steel. Fittings used in the water, low pressure hydraulic and low pressure air lines may have ends of the National Pipe Thread (NPT) type. The external water and air line (inlet) bosses shall be of the NPT type. Teflon tape is acceptable for sealing threaded joints.



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3.3.2.2 Flexible hose. Black oxide finish as corrosion preventative measure on hose fittings and hose ends is acceptable.

3.3.3 Hydraulic and pneumatic components. Hydraulic and pneumatic systems components shall comply with the requirements specified herein.

3.3.3.1 Oil reservoir. A stainless steel hydraulic fluid reservoir shall be provided and mounted within the stand enclosure. The reservoir shall have adequate capacity to supply at least 25 gallons of fluid to the system plus additional space for a three-gallon expansion above the normal full level. The reservoir shall conform for obstruction to flow, structural strength, baffles, fluid connections, sump, drain, vent, vent filter, vent line, return line inlet, filler unit, finish, and filling time. Adequate clearance around the filler opening shall be provided to allow rapid filling from one-gallon cans while the stand is in operation. The reservoir shall include systems for venting and for pressurizing with nitrogen to 0.5 psig maximum. Each of the two conditions shall be manually selectable from the panel of the stand. A low liquid level switch shall be provided that will shut down the stand when the liquid level is below the one-half full point. The reservoir fluid connections may be pipe thread (NPT) connection.

3.3.3.1.1 Sump filter. The sump filter (strainer) shall consist of a 60 mesh recleanable strainer. It may be located inside the reservoir as shown in Figure 1, or outside the reservoir inline immediately below the suction shutoff valve (6).

3.3.3.1.2 Liquid level gauge. The liquid level gauge shall be of the mechanical type**.** The gauge shall indicate the liquid level in the 1/4, 1/2, 3/4 and the full level positions. It is unnecessary to indicate the level in gallons



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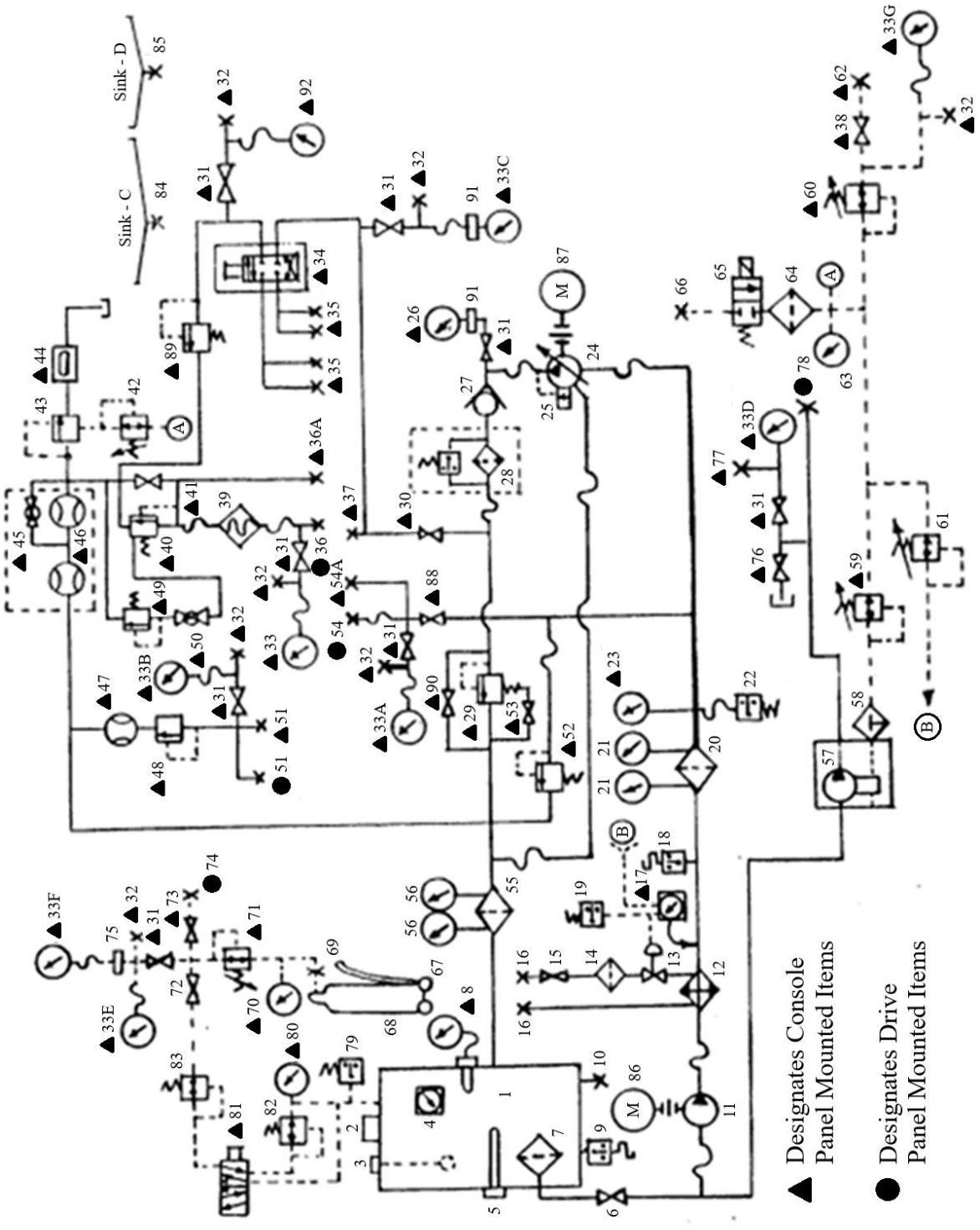



**FIGURE 1 HYDRAULIC AND PNEUMATIC SCHEMATIC**



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Key to FIGURE 1.

| | | | |
|---|---|---|---|
| 1. | Reservoir | 33E. | Test gauge, 0 to 3000 psig |
| 2. | Filler cap | 33F. | Test gauge, 0 to 800 psig |
| 3. | Low level switch | 33G. | Test gauge, 0 to 160 psig |
| 4. | Liquid level indicator | 34. | Cycling valve |
| 5. | Immersion Heater | 35. | Cycling ports |
| 6. | Suction shutoff valve | 36. | Test pump discharge port |
| 7. | Sump filter, 60 mesh | 36A. | Return port |
| 8. | Reservoir temperature indicator gauge, 40 to 240 °F | 37. | High pressure outlet port |
| 9. | Temperature switch | 38. | Air supply shutoff valve |
| 10. | Reservoir drain port | 39. | Ripple filter |
| 11. | Boost pump | 40. | Discharge high pressure control |
| 12. | Heat exchanger | 41. | Discharge throttling valve |
| 13. | Temperature control valve | 42. | Relieve type regulator |
| 14. | Strainer | 43. | Relief valve |
| 15. | Valve | 44. | Flowmeter over pressure sight flow indicator |
| 16. | Water ports | 45. | Flowmeter bypass valve |
| 17. | Fluid temperature controller | 46. | Discharge flowmeters, 0.45 to 35 gpm |
| 18. | Temperature switch | 47. | Case drain flowmeter, 0.13 to 1.66 gpm |
| 19. | Pressure switch | 48. | Case drain back pressure regulator valve |
| 20. | Filter, 10 micron nominal | 49. | Discharge low pressure control |
| 21. | Filter pressure gauges, 0 to 200 psig | 50. | Discharge pressure cycling valve |
| 22. | Pressure system boost switch | 51. | Case drain return ports |
| 23. | Boost supply pressure gauge, 0 to 200 psig | 52. | Boost pressure relief valve |
| 24. | High pressure pump | 53. | High pressure system dump valve |
| 25. | Pump pressure compensator | 54. | Test pump inlet port |
| 26. | Pressure gauge, 0 to 6000 psig | 54A. | Boost supply outlet port |
| 27. | Check valve | 55. | Filter, 10 micron nominal |
| 28. | High pressure filter, MIL-F-27656 | 56. | Filter pressure gauges, 0 to 200 psig |
| 29. | High pressure system relief valve | 57. | Air operated pump |
| 30. | High pressure outlet shutoff valve | 58. | Air line lubricator |
| 31. | Gauge shutoff valves | 59. | Air operated pump regulator |
| 32. | Cage calibration ports | 60. | Air supply pressure regulator valve |
| 33. | Test gauge, 0 to 6000 psig | 61. | Air supply pressure regulator |
| 33A. | Test gauge, 0 to 150 psig | 62. | Air supply outlet port |
| 33B. | Test gauge, 0 to 160 psig | 63. | Pressure gauge, 0 to 200 psig |
| 33C. | Test gauge, 0 to 6000 psig | 64. | Air supply moisture separator filter |
| 33D. | Test gauge, 0 to 30000 psig | 65. | Solenoid valve |



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Key to FIGURE 1 – Continued.

| | |
|---|---|
| 66. | Shop air inlet port |
| 67. | Cylinder cart |
| 68. | Nitrogen cylinder |
| 69. | Cylinder outlet |
| 70. | Nitrogen pressure gauge, 0 to 4000 psig |
| 71. | Nitrogen supply pressure regulator |
| 72 | Valve |
| 73 | Nitrogen test outlet shutoff valve |
| 74 | Nitrogen test outlet port |
| 75 | Gauge saver |
| 76 | Static test bleed valve |
| 77 | Calibration port |
| 78 | Static test outlet port |
| 79 | Reservoir pressure switch |
| 80 | Reservoir pressure gauge, 0 to 3 psig |
| 81 | Push pull valve, reservoir pressure/vent |
| 82 | Nitrogen regulator |
| 83 | Nitrogen regulator |
| 84 | Sink drain port |
| 85 | Sink drain port |
| 86 | Motor |
| 87 | Motor |
| 88 | Valve |
| 89 | Back pressure regulator |
| 90 | Bypass valve |
| 91 | Pulsation dampener |
| 92 | Test gauge, 0 to 300 psig |



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3.3.3.2 Low pressure boost pump. A hydraulic boost pump shall be provided. The boost pump shall supply fluid continuously to the high pressure pump with pressures recommended by the high pressure pump manufacturers. An interlock system shall be incorporated which will shut the test stand down if the boost pressure falls below the recommended pressures. The boost pump shall be a positive displacement type, with relief valve protection. The boost pump shall also provide boost pressures for the pump test circuit.

3.3.3.2.1 Pressure gauge. A 0 to 200 psig service type pressure gauge shall be provided downstream of the hydraulic boost pump. The gauge shall be mounted on the test stand panel.

3.3.3.3 Filters. The pump discharge filters shall be in accordance with MIL-F-27656 filter elements. Each filter assembly shall contain a suitable differential pressure gauge, (or a differential pressure switch with warning light), to indicate the need for element replacement or cleaning. There shall be no bypass valves included. Sufficient clearance space shall be provided for the replacement of filter elements when installed on the test stand. The filter assemblies shall provide drain and vent valves as required. If practicable, the indicators for the filter shall be grouped to facilitate the observation of incipient plugging. It is not necessary that these indicators be located on the main instrument panel but can be located on an easily accessible subpanel within the enclosure.

3.3.3.4 Heat exchanger. Cooling of test fluid shall be accomplished through a heat exchanger using water where the desired oil temperature is between 100 and 160 °F. The heat exchanger shall be sized to provide the required cooling at al1 times when operating at an ambient temperature of 100 °F using cooling water at 80 °F.

3.3.3.4.1 Capacity. The heat exchanger shall have sufficient capacity to maintain a 110 °F oil temperature when total horsepower capacity of the test stand is dissipated into heating the oil through relief valve or throttling action of the control valves over a period of one hour.

3.3.3.4.2 Temperature control. Automatic controls shall be provided for cooling the test fluid and shall maintain the set temperature within ±2 °F and shall be adjustable to permit setting control temperature at any point between 100 and 160 °F. The control shall also indicate actual oil temperature and shall be located on the control panel convenient and visible to the operator's position. No manual control valves shall be required in addition to the automatic control. Shutoff valves shall be interlocked to operate automatically with the control of the test equipment motors. The temperature of the cooling medium shall**.** be as specified under 3.3.3.4; however, the control shall operate successfully with cooling water of a lower temperature than specified. The temperature control shall maintain the temperature within the required limits during all conditions normally encountered during the test. The response of the control shall be fast enough to maintain the temperature within the required limits during transient conditions to permit uninterrupted testing. All valve parts coming in contact with the cooling water shall be of bronze or other corrosion-resistant material. The valve locations shall afford easy access for adjustment or repairs. A flow dividing valve may be used in the return line to control temperature, but it shall be designed so that it cannot block the return line under any condition.



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3.3.3.4.3 Cooling water filter. A removable screen water filter shall be placed in the cooling water supply line immediately preceding the oil temperature control valve. The filter shall have a minimum capacity of 25 percent greater than that of the heat exchanger cooling coil. All filter parts shall be made of brass, bronze, or an equally corrosion-resistant material. The filter screen shall have not less than 60 or more than 80 mesh per inch and shall have reinforced edges.

3.3.3.4.4 Manual water control valve. A slow closing globe or gate type valve shall be installed in the cooling water supply line. The valve hand wheel shall be easily accessible and when in the closed position shall shut off the water supply to the filter, thermal control valve**,** and the cooling coil.

3.3.3.5 Sump drain. The drain must not have a return system to avoid any contamination or pollution possibilities.

3.3.3.6 Heaters. Heating of the test fluid to raise its temperature shall be by the use of low watt density immersion heaters. Electrical immersion heaters used for heating test fluid must be UL approved for this purpose and shall be designed *so* the heating elements are submerged a minimum of six inches below the surface of the fluid at all times. The fluid well in which the heating elements are installed shall be equipped with a low fluid level switch that shuts the test standoff automatically when the fluid level is less than six inches above the heating elements. The fluid well shall also be equipped with a UL approved overtemperature switch that shuts the heater off automatically when the test fluid temperature reaches 180 °F.

3.3.3.7 Relief valves. Commercial type relief valves may be used and shall be selected for ease of adjustment and maintenance, long life, and suitability for the purpose. The main relief valves shall be adjustable between 500 and 5000 psi.

3.3.3.7.1 Bypass valve. A panel mounted throttling valve (bypass) installed parallel to the high pressure (main) system relief valve, or an equivalent alternate provision, shall be provided to allow test fluid warmup without upsetting the high pressure system relief valve setting.

3.3.3.8 Hydraulic pump. The stand shall be equipped with a hydraulic pump with a capacity of 30 gpm at 3750 psi and reduced flow at 5000 psi. The pump shall be of the variable volume, axial piston type, with pressure balanced piston shoes. The pump shall be equipped with a stroke control to permit regulation of maximum volume delivered by the pump. The pump shall also be equipped with an adjustable automatic pressure compensator control capable of regulating pressure at a constant value with flow varying from minimum flow setting to full flow. Pump pressure pulsations shall not exceed plus or minus 75 psi at 1000 psi, plus or minus 100 psi at 2000 psi, plus or minus 125 psi at 3000 psi, and plus or minus 250 psi at 5000 psi. The pump shall have a minimum life expectancy of 1,000 hours of continuous duty at full rated load. The pump shall be capable of meeting all requirements in 4.5.2.2.



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3.3.3.8.1 Pressure gauge. A 0 to 6000 psig service type pressure gauge shall be provided downstream of the hydraulic pump for aid in setting up the system operating pressure. The gauge shall be mounted on the test panel and shall be equipped with a shutoff valve and hydraulic dampener.

3.3.3.8.2 Pump control handles. The handles for the pumps maximum volume control and automatic pressure compensator control shall be located on the front side of the test stand so as to permit the test stand operator to adjust the corresponding flow and pressure while observing the test stand flowmeter and pressure gauges.

3.3.3.9 High pressure pneumatic supply. A static pressure test circuit capable of providing pneumatic pressures up to 3000 psig shall be provided. The source shall be a suitable rechargeable nitrogen cylinder which is capable of being easily connected to the test stand. The cylinder shall be mounted on a suitable cart for transportation and shall be similar to Interstate Commerce Commission Series 3AA cylinder with a volume of 2630 cubic inches. The cylinder and cart shall be furnished as part of the test stand. Pressure regulators shall be of sufficient accuracy and sensitivity to control pressure settings within ±1 percent of present values within the range of 100 to 3000 psig. Two gauges with 0.25 percent foot per second (ft/s) accuracy to cover the range of 100 to 3000 psig shall be supplied.

3.3.3.10 Air operated pump. The stand shall be provided with an air operated pump to supply high hydraulic test pressures. The pump shall be capable of supplying the stand with 25,000 psi hydraulic test pressure and shall be powered by an air motor. The pump shall be capable of delivering the 25,000 psi when 80 psi shop air is supplied to the pump motor. The stand shall be provided with the necessary valves, filters, regulators, and other hardware required to provide the stand with dependable and long life operation of the pump.

3.3.3.11 Air connection. The stand shall be provided with a shop air connection circuit to supply filtered pressure regulated air to the stand air components and to a panel mounted low pressure air test outlet. The circuit shall include a 0.375 in NPT air supply inlet, solenoid shutoff valve, moisture separator filter, supply pressure gauge, pressure regulators, shutoff valve, test gauge, gauge calibration port and test port.

3.3.3.12 Water connections. Water and drain connections for the heat exchanger shall be directed to and from the stand through suitable NPT outlets located near the rear external surface of the stand.

3.3.3.13 Shield. A transparent, protective, impact resistant shield shall be provided to enclose the mounting pad and item-under-test section of the drive console to minimize danger to the test stand operator and to allow meeting the specified noise level requirement. An aircraft pump will be loaned to the contractor to enable the design of the shield to meet the physical and noise level requirements.

3.3.3.14 Cycling ports. Dual pressure and dual return cycling ports shall be provided as part of the dynamic test circuit.



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3.3.3.15 Return pressure regulators. Test circuits shall include regulators in the return lines to provide pressure regulation as follows:

| Test circuit | Range | Control accuracy |
|---|---|---|
| Pump discharge, high | 500-3200 psig | ±25 psi |
| Pump discharge, low | 100-3000 psig | ±25 psi |
| Pump case drain | 0-160 psig | ±5 psi |
| Dynamic | 0-200 psig | ±5 psi |

The regulators shall be adjustable from the panel of the test stand. The test pump discharge regulators shall be provided with a panel mounted selector for cycling the pump load from high pressure to low pressure and back to high pressure, etc.

3.3.3.16 Ripple filter. A high pressure hydraulic ripple filter, ACL-Filco part number S1002-20-5-3750S, shall be provided in the test pump return line to eliminate the transmission of pump ripple in the measuring instructions of the test stand. An equivalent ripple filter which will eliminate pump ripple transmission to the measuring instruments and allow the test stand to meet its end performance requirements will also be acceptable.

3.3.3.17 Shutoff and control valves. The 90 degree handle motion does not apply-to shut off and control valves of this test stand. More travel is necessary to give proper control.

3.3.4 Instrumentation. Hydraulic and pneumatic systems instrumentation shall comply with the requirements specified herein.

3.3.4.1 Pressure indicating gauges and instruments. All pressure gauges, temperature indicators, and dial indicating instruments shall be particularly suited for hydraulic test service. All pressure indicating gauges shall incorporate the best quality bourdon tube elements which exhibit long endurance, low hysteresis, and high fatigue resistance. All gauge movements shall be designed to withstand continuous high rates to vibrations and pulsations. Each pressure gauge shall indicate values within plus or minus one-half of one percent of the full scale range of the gauge. All hydraulic and pneumatic pressure gauges shall have blow-out plugs. All pressure test gauges shall be of six inches dial size.

3.3.4.2 Instrument mounting. All indicating instrument dials and sight tubes shall be flush panel mounted. Hydraulic pressure gauges shall be mounted by screws in tapped holes to permit removal from the front of the panel by one man. All hydraulic pressure gauges shall be connected to the panel with a flexible hose suitable for the purpose and of sufficient length to permit removal and installation from the front of the test stand.

3.3.4.2.1 Hydraulic dampeners. Hydraulic dampeners (snubbers) shall be installed to protect all pressure gauges on the output side of the test stand's hydraulic pump.



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3.3.4.3 Gauge valves. All pressure gauges shall be equipped with needle valves located as close as practicable to the gauge. All gauges subject to pressure beyond their maximum range shall be protected with standard commercial type automatic gauge shutoff valves. Automatic gauge shutoff valves shall be of positive action which will allow absolutely no leakage of fluid to the gauge. A suitable in-line filter shall be provided in the circuit immediately ahead of the automatic gauge shutoff valves.

3.3.4.4 Cage calibration outlets. Each pressure test gauge shall be provided with a calibration outlet (bulkhead fitting) with end conforming to SAE-AS4395, style E, size 4 with mating pressure cap to permit gauge calibration without removal of the gauge. The outlets shall be located on the panel such that they are convenient to the operator's position and provide maximum safety to the operator. Each outlet shall be labeled according to function.

3.3.4.5 Non-test service type gauges. Except as otherwise specified herein, non-test service type reference gauges need not meet the accuracy requirement nor be equipped with hydraulic dampeners, calibration ports and shutoff valves as specified in 3.3.4.1 through 3.3.4.4 above.

3.3.4.6 Oil reservoir temperature gauge. The temperature gauge used to indicate the oil temperatures shall be of the vapor tension, immersion type. The gauge dial shall be graduated to read from 40 to 200 °F and accuracy throughout the range shall be within ±2 percent. A red warning line shall be placed at the 160 °F mark to indicate the standard high temperature limit. The temperature gauge shall be control console panel mounted.

3.3.4.7 Flowmeter. Flowmeters shall be provided to measure the flow of the main circuit and pump case drain. The main flow circuit shall consist of a two-stage flow meter with ranges of 0.45 to 4.5 gpm and 4.5 to 35 gpm. A selector valve shall be provided to direct the fluid to the desired flow meter. The case drain flowmeter shall have a range of 0.13 to 1.66 gpm and be 250 MM in length. The flowmeter shall be calibrated for use with oil conforming to specification MIL-PRF-5606. The flowmeters shall be easily readable. The two-stage main flowmeters shall have meter tubes and scales incorporating logarithm characteristics to provide uniform percentage readability at all points on the scale. The flowmeters shall be mounted in a single shell or enclosure and shall be readily accessible for cleaning without removal of glass tubes from the stand. A relief valve shall be provided to protect the main flowmeters from excessive pressures and connected to a return line to the reservoir. This line shall have a sight flow indicator to reveal any leakage past the relief valve. Indicating accuracy of the flowmeters shall be within ±l percent throughout the range and through a temperature range between 100 °F and 160 °F. The flowmeters shall have rod guides for the floats or come other method of discharging static electricity.

3.3.4.8 Handles. All control handles, knobs and other devices shall be maintained below 110°F.



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3.4 Electric motors, controls and electrical circuits. The test stand shall contain installed electrical equipment as required to meet the requirements specified herein.

3.4.1 General. The electrical system shall conform to the applicable requirements of American Standard C-50, NEMA Standard MG-1, and National Fire Protection Association Code No. 70. All wiring and electrical components to be constructed from fungus resistant material per MIL-HDBK-454.

3.4.1.1 General requirements. The test stand shall have all components electrostatically bonded together with an electrical continuity of 25 ohms (or less) resistance i n accordance with National Electric Code (NFPA Code No. 70). Electrical continuity test shall be performed throughout the test stand to ensure compliance with this requirement. A plainly marked grounding connection shall be provided on the test stand for connection to the facilities grounding system. Unless otherwise specified in the ordering data (see 6.2) the test stand shall operate from a 440-volt, 3 phase, 60 Hertz AC power supply. If the test stand is to be built for power other than 440-volt, 3 phase, 60 Hertz AC.

3.4.1.2 AC control. In addition to reduced voltage starters for motors of 25 horsepower and above part winding starting of such motors may be used where appropriate.

3.4.2 Variable speed drive. The test stand shall be provided with a reversible and variable speed drive for the pump test pad. The drive motor shall be a variable speed electric motor. Drive shall be direct (no belt or mechanical gears will be acceptable). The drive output speed shall be adjustable by means of a hand wheel located convenient to the test stand operator. The speed range of the drive shall be from 500 to 5000 revolutions per minute (rpm). The maximum constant torque output shall be at least 1000 in-pounds at speeds up to 3750 rpm. The maximum constant horsepower output shall be at least 60 horsepower at speeds above 3750 rpm. Automatic speed regulation of 0.5 percent ft/s shall be provided. In the speed range of 2700 to 5000 rpm, maximum transients upon sudden load shall not exceed three percent of the set speed and shall be effectively dampened in less than one second.

3.4.3 Pump test mounting pad. The stand shall be provided with a hydraulic pump test mounting pad located convenient to the operator and the pump test panel outlets. Drive pads to which units will be attached shall be designed so that the mounting pad and spline adapters will be fully removable and interchangeable with other sizes of splines and pads. The pads and splines provided shall be in accordance with standards AND20001 as applicable to aircraft engine hydraulic pumps.

3.4.4 Speed indicator. A tachometer indicator with a range of 0-5000 rpm shall be provided on the front panel. The tachometer indicator shall be connected with a speed sensing device on the variable speed drive. The indicator shall indicate pump test pad speed and have an accuracy of ±1 percent of full scale.



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3.4.5 Timer. An elapsed digital time meter shall be provided. The meter shall be panel mounted, have a range of 0 to 9999 seconds, and have a reset button.

3.4.6 DC power supply. A panel mounted DC power supply capable of being varied between 18 and 30-volts shall be supplied. The power supply shall be capable of supplying 0-15 amperes. The voltage shall be manually adjustable over the range specified. The power supply shall include the necessary voltmeters, ammeters, pilot lights , control switches and terminal connections necessary to provide the electrical supply necessary for the operation of various electrically actuated hydraulic components. The voltage shall remain within ±l percent of the set value for a maximum load variation.

3.4.7 AC power supply. A panel mounted 115-volt, 400 cycle, single phase, 500 watt power supply capable of being varied between 110 and 120-volts shall be supplied. The frequency shall be adjustable between 360 cycles and 440 cycles per second (cps). Regulation of the voltage and frequency shall be within ±1 percent. The power supply shall include the necessary voltmeters, ammeters, pilot lights, control switches necessary to provide the electrical supply necessary for the operation of various electrically operated hydraulic components.

3.4.8 Pump suction pressure switch. Pump suction line shall have a pressure control power circuit breaker switch which shall operate to stop the motor when the pressure at the pump suction port falls below the recommended minimum value. A panel light shall indicate motor stoppage caused by actuation of the circuit breaker switch.

3.4.9 Lighting. The volume flowmeters shall be illuminated by lights of ample wattage. The lighting circuit shall be powered by 110-volt AC, 60 cycles, single phase current. Other instrumentation shall be selected in so far as practicable to eliminate need for integral lights.

3.4.10 Torque meter system. A torque meter system shall be provided for measuring the torque of pumps driven by the variable speed drive. Indicating range shall be at least 0 to 1000 inch-pounds. Overall accuracy shall be ±3 percent of full range throughout the indication range of 100 to 1000 inch-pounds. The indicator shall be mounted on the control console panel.

3.5 Reliability. Each test stand shall have a minimum Mean-Time-Between-Failures (MTBF) of 250 hours. Failure is any malfunction which requires test stand "downtime," whether immediate or deferred, for adjustment, repair or replacement of any component. A malfunction is defined as any condition which prevents the test stand from operating within the performance specified herein.

3.6 Maintainability. The test stand shall be designed to assure maximum maintainability consistent with the requirements specified herein and the requirements of MIL-HDBK-470. The Mean-Time-To-Repair (MTTR), assuming spare parts, tools and test equipment are on hand, shall not exceed four hours. The maximum time required for any repair/replacement of any components or assembly shall not exceed six hours.



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3.7 Safety. This test stand shall be designed to assure maximum safety, consistent with the requirements specified herein, and that design characteristic, environmental factors, personnel errors, procedural deficiencies or system/subsystem/component failure or malfunction will not result in injury to personnel or damage to the test stand and/or other property. The contractor shall establish a program of safety analysis and shall implement this program. Guidelines called out in MIL-STD-882 shall be followed. While this program will be established and conducted exclusively by the contractor, complete records shall be kept and be available to Government personnel.

3.7.1 Safety devices. Automatic "shut down" safety device shall be included in the test stand to preclude hazardous conditions resulting from test fluids over-temperature and/or rupture or leak in fluid lines. For example, a temperature shut down device shall be placed downstream from the heat exchanger so that the stand will be shut down when the oil temperature of this location approaches 180 °F. Consideration to preclude damage to equipment and items under test shall be included; however, first consideration shall be given to personnel safety.

3.7.2 Hazard analysis. The contractor shall perform an operational hazard analysis on the test stand and present the results in a written report to be received at least two weeks before first article demonstration.

3.8 Audio noise level. The test stand shall be designed so that the noise level produced by the test stand and the component under test and throughout the operational range shall not exceed 85 decibels on the A-weighted scale in any frequency range between 300 and 4800 cps at any point outside of the test stand, The audio noise level will be tested per paragraph 4.5.3.

3.9 Finishes and protective coatings. Cleaning, painting, plating and anodic films shall be in accordance with MIL-HDBK-808. Test stand finish color shall be as follows:

a. Exterior surface of the test stands, except where stainless is specified herein, shall be semi-gloss gray number 26373 conforming to SAE-AMS-STD-595.

b. Interior of the test stands shall be semi-gloss white number 27886 conforming to SAE-AMS-STD-595. Exceptions to the white finish are areas requiring acoustical-treatment to meet noise level requirements specified herein.

c. Cadmium plating usage shall be prohibited on components of the hydraulic and pneumatic test circuits and on test stand components that can come into contact with aircraft items to be tested.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Classification of test. The inspection and testing of the stand shall be classified as follows:

a. Preproduction tests. (See 4.5)

b. Acceptance tests. (See 4.3)

4.3 Acceptance tests. Acceptance tests shall consist of individual tests.

4.3.1 Individual tests. Each test stand shall be subjected to the following tests as described under 4.4, test methods, of this specification.

a. Examination of product.

b. Preliminary run-in and calibration test.

c. Constant flow operation.

d. Test stand operation.

e. Gauge tests.

f. Variable speed drive tests.

g. Electrical power supplies tests.

h. Static pressure systems tests.

i. Low pressure hydraulic boost supply tests.

j. Subsystems tests.

k. Safety devices and interlock tests.

l. Functional tests.

4.4 Test methods.

4.4.1 Examination of product. Each test stand shall be inspected to determine compliance with requirements specified herein with respect to parts, materials, finish, workmanship, and markings.

4.4.2 Preliminary run-in and calibration test. Any preliminary run-in of the test stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the test stand for acceptance tests. A certified log of all preliminary running time, calibration data, etc., shall be submitted to the procuring activity.

4.4.3 Constant flow operation. The constant flow operation of the test stand shall be checked at 800, 1500, and 3000 psi at each rate of flow of 2, 4, 6, 20, and 30 gpm and 5000 psi at 15 gpm. The flow shall be set by the main pump's maximum volume control. The flow shall stabilize within ±1/2 gpm of the set flow. The test stand shall then be operated at 15 gpm and 5000 psi for one hour and inspected for evidence of leakage of fluid.



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4.4.4 Test stand operation. The operation of the test stand shall be checked at 1500 and 3000 psi at each rate of flow of 5 and 30 gpm and with compensation set at 1500 and 3000 psi, respectively. The hydraulic pump shall put out full *5* and 30 gpm flow at all loading pressures below 1500 and 3000 psi, respectively, but flow shall fall to zero as soon as the loading pressure reaches the compensation pressure setting.

4.4.5 Gauge tests. All pressure gauges, flowmeters, speed and torque indicators and other electrical meters shall be cycled at least 10 times and then rechecked for calibration.

4.4.6 Variable speed drive tests. The variable speed drive system shall be operated in both clockwise (CW) and counterclockwise (CCW) rotation with maximum horsepower loads at 3750 rpm and at 5000 rpm, and with lesser loads at a minimum of six speeds throughout the specified speed range to demonstrate acceptable operation of the drive, its controls and the indicating meters. For at least one of the above conditions in both CW and CCW rotation within the 2700 and 5000 rpm speed range, the load shall be suddenly varied at least six times each to demonstrate acceptable transient response and recovery. The loading devices and transient response/recording equipment shall be contractor furnished.

4.4.7 Electrical power supplier tests. The AC and DC power supplies shall be operated at maximum load conditions and at a minimum of six load conditions throughout the respective specified voltage, amperage and frequency ranges to demonstrate acceptable operation of the power supplies, the controls and the indicating instruments. The loading device(s) shall be contractor furnished.

4.4.8 Static pressure systems tests. The static high pressure pneumatic and hydraulic supply systems shall be operated to maximum pressure specified and to at least six pressure levels throughout the respective specified pressure ranges to demonstrate acceptable operation of the systems, the controls and the indicating instruments.

4.4.9 Low pressure hydraulic boost supply tests. The low pressure hydraulic boost supply circuit shall be operated at a minimum of six pressure and flow combinations throughout the specified range to demonstrate acceptable operation of the supply, its controls and the indicating instruments. Any external device(s) required to achieve the test shall be contractor furnished.

4.4.10 Subsystems test. The test stand shall be operated to demonstrate acceptable operation of the reservoir venting and pressurization systems and the hydraulic fluid temperature control systems and associated instruments. Operation at maximum temperature and at six other temperatures throughout the specified range shall be included.



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4.4.11 Safety devices and interlock tests. The following shall be tested to demonstrate their proper operation:

a. All safety devices.

b. Test stand grounding.

c. Automatic gauge shutoff valves.

d. All interlocks.

e. Emergency switches.

f. Low fluid level switches.

g. Overtemperature switches.

h. Flowmeter relief valves.

4.4.12 Functional tests. The test stand shall be operated through its specified capacity and pressure range by the performance of actual test of two aircraft hydraulic components. A test shall be performed to demonstrate test stand capabilities for testing aircraft hydraulic pumps. A standard engine-driven pump shall be selected and pressure, torque and flow tests performed as required by the pump manufacturer. The necessary adapters that will enable the pump to interface with the stand shall be furnished by the contractor.

4.5 Preproduction tests.

4.5.1 Preproduction test sample. The preproduction test sample shall consist of one test stand representative of the production equipment. Preproduction tests shall be performed at the contractor's plant under supervision of the procuring activity.

4.5.2 Endurance test.

4.5.2.1 The test stand shall be subjected to 350 hours of endurance testing Readings shall be recorded every hour of all gauges, meters, and instruments on the test stand. The stand shall be subjected to the following 350 hours of endurance testing. All test periods of less than three hours duration will not be recorded as part of the endurance test. Loading devices shall be contractor furnished.

a. A test program, within the power and heat exchanger capacity of the test stand, to provide a maximum combined degree of loading of the test stand drive and test stand hydraulic system.

b. Shall include a maximum loading for 25 percent of time of each system, i.e., test stand drive and test stand hydraulic system.



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4.5.2.2 During the endurance test, the flow shall be varied instantaneously from the preset flow to zero flow and immediately returned again to the preset flow. This cycle shall be accomplished 1000 times during each 50 hours of testing. Any damage to the pump, pump compensator, flow indicator, gearing, or other components shall be cause for rejection. The load on the drive shall be suddenly varied from the preset load to no-load and immediately returned again to the preset load. This cycle shall be accomplished 1000 times during 50 hours of testing. Any damage to the mounting pad, splines, instruments, bearings or other components shall be cause for rejection.

4.5.2.3 The test as specified in 4.3.1 shall be repeated. These data shall be compared with those recorded in 4.5.2.1. Results shall indicate that the stand is in excellent condition and that no failure or undue wear has resulted from the endurance test.

4.5.3 Audio noise level. The noise level produced by the test stand and the component under test during testing throughout the operational range shall not exceed 85 decibels on the A-weighted scale in any frequency range between 300 and 4800 cps at any point outside of the test stand. Noise level measurements shall be made while the test stand is operated through the speed range at full rated load. During the test, microphones shall be located one foot from the stand at any point.

4.5.4 Maintainability. The contractor shall show proof that he implemented the various maintainability programs specified in MIL-HDBK-470 and related documents.

4.5.5 Safety demonstration. The test stand shall be inspected and functionally checked to ascertain that safety requirements have been incorporated in the design and manufacture of the stand. Potential hazards shall be identified, the safety device or method shall be shown, and an assessment of adequacy shall be made.

4.5.6 Reliability demonstration. The contractor shall demonstrate the test stand meets, or exceeds, the specified reliability. This may be shown by any combination of test, calculations and established MTBF data for components.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)



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6.1 Intended use. This test stand is intended to test the hydraulic components used in the F-4 aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Level of packaging and packing required.

c. The electrical power the test stand will be connected to for operation if other than 440-volt, 3 phase, 60 hertz AC power (see 3.4.1.1).

d. Aircraft components to be loaned government furnished equipment (GFE) to the contractor for preproduction and acceptance tests (see 4.4.12).

e. The quantity of preproduction samples (see 6.3.3).

6.3 Instructions for preproduction samples.

6.3.1 Contractor's demonstration tests. The Government inspector and the procuring activity needs to be advised when tests are to be conducted so that a representative may be designated to witness the test when so desired. The contractor needs to submit all data collected in conducting these tests to the procuring agency for review and approval.

6.3.2 Preproduction sample approval. Approval of the preproduction sample needs to be by procuring activity upon satisfactory completion of the preproduction tests. No production equipment needs to be delivered prior to the approval of the preproduction model.

6.3.3 Preproduction samples. Unless otherwise specified in the order or contract, the first manufactured unit of each type, as distinguished by electrical power from which it will operate (see 3.4.1.1 and 6.2.c), needs to be subjected to the preproduction tests of 4.5.

6.4 Subject term (key word) listing.

Dynamic test circuit
Maximum-fluid velocity
Hydraulic boost pump
Tachometer
Counterclockwise

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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Preparing activity:
Air Force - 184

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